Lecture 7(a): Flexibility baseline

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Overview of Baseline in Flexibility Quantification

Baseline Functionality

Baselines estimate electricity use without flexibility activation, serving as a reference for measuring flexibility delivered.

Applications of Baselines

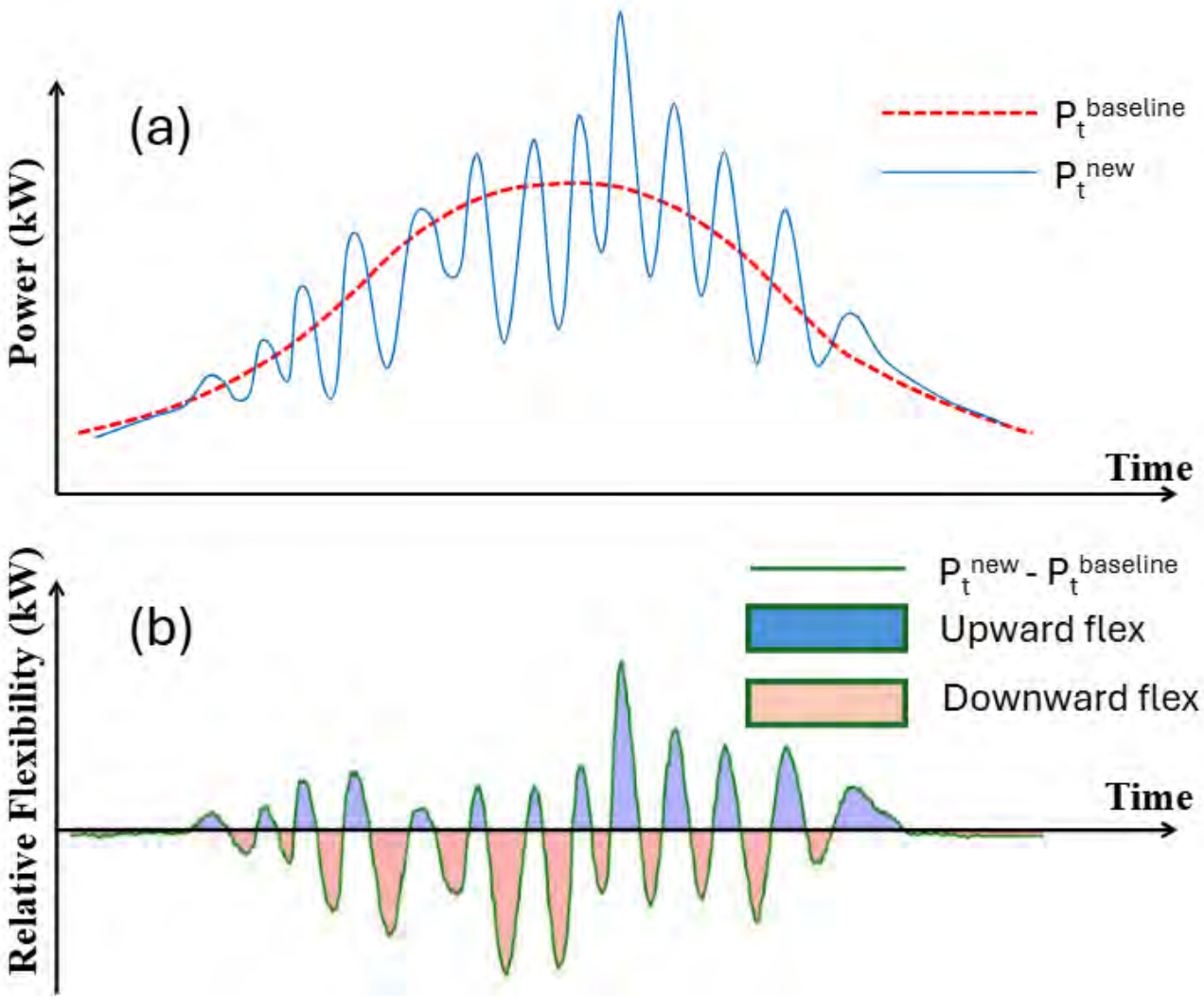
Baselines are crucial for demand response, settlement, performance verification, and ensuring measurable energy changes.

Challenges in Baseline Calculation

Variability in behavior, weather, asset cycling, and market rules complicate accurate baseline computation.

What is a baseline?

NAESB definition: A Baseline is an estimate of the electricity that would have been consumed by a Demand Resource in the absence of a Demand Response Event.



Source: https://www.nyiso.com/documents/20142/1410950/EnerNOC_Baseline_Presentation_09162010.pdf/73bf3f52-1466-f0c5-07c7-d0cb48999e2d

Need for Baseline

Importance of Baseline in Flexibility Measurement

Role of Baselines in Measurement

Baselines provide a reference point to measure actual load or generation changes during flexibility events accurately.

Ensuring Fair Compensation

Baselines ensure compensation rewards true behavioral changes, preventing gaming and aligning payments with actual flexibility.

Supporting Grid Stability and Planning

Accurate baselines improve forecasting, operational planning, and maintain grid stability through reliable flexibility data.

Regulatory Compliance and Transparency

Baselines promote transparency and fairness among resources, supporting compliance and pay-for-performance market models.

Problems Associated with Baseline Models

Inaccuracy from Variable Conditions

Baseline models struggle with inaccuracy due to unpredictable user behavior, weather changes, and operational variability.

Risk of Manipulation

Participants can game baseline models by altering consumption to unfairly influence future baseline values.

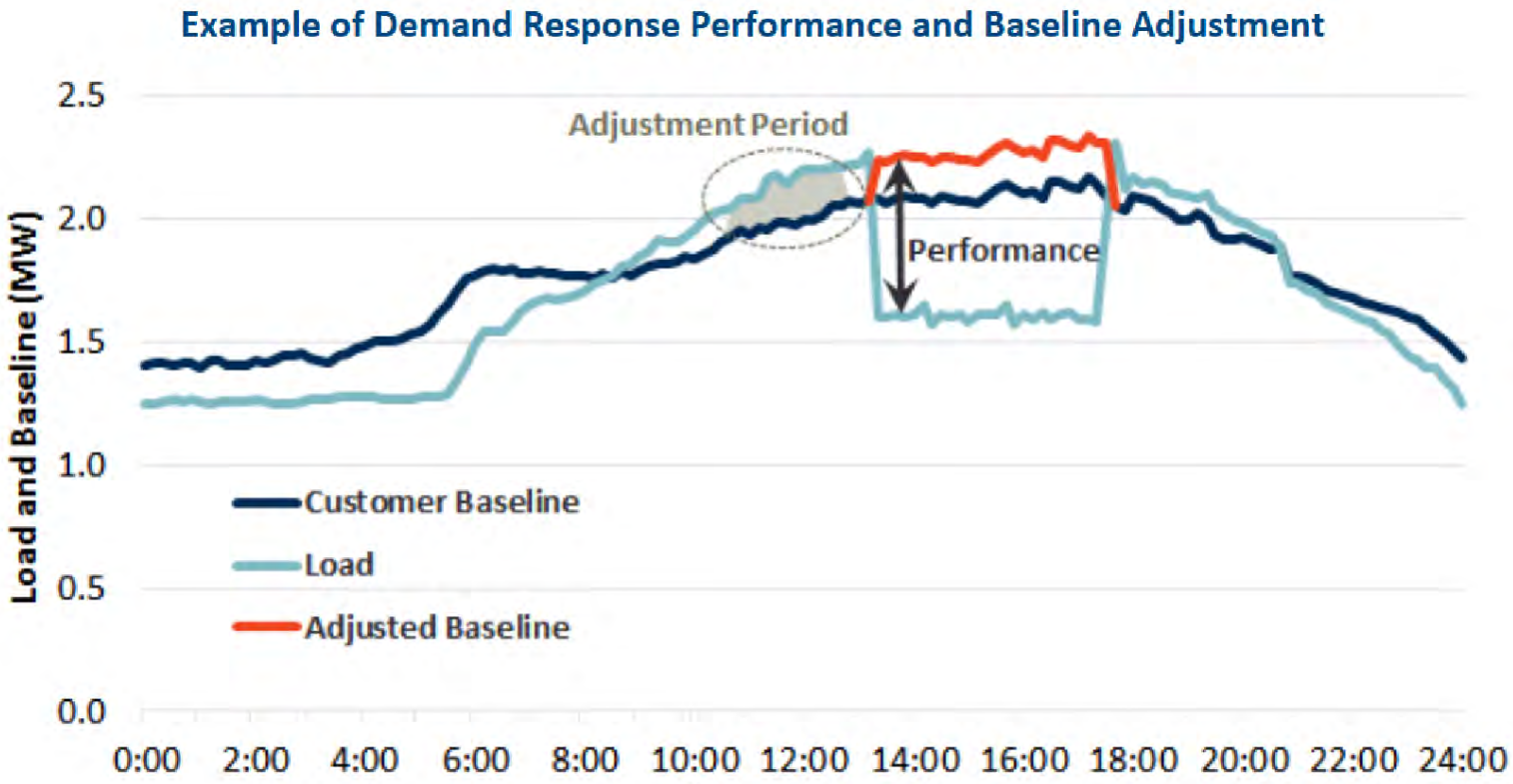
Model Bias and Complexity

Baselines may bias certain consumer types and face complexity when aggregating diverse assets with fast-changing loads.

Need for Real-Time Alternatives

Conventional baselines fail in inverter-dominated grids, highlighting the need for real-time performance-based models.

Baseline adjustment to manipulation



Source: Adapted from an ISO New England presentation, *Introduction to Demand Response in ISO-NE Markets*, May, 2015.

Source: https://www.aemc.gov.au/sites/default/files/content/9207cd67-c244-46eb-9af4-9885822cefbe/Final-AEMC-DR-Report_International-Review-of-Demand-Response-Mechanisms.pdf

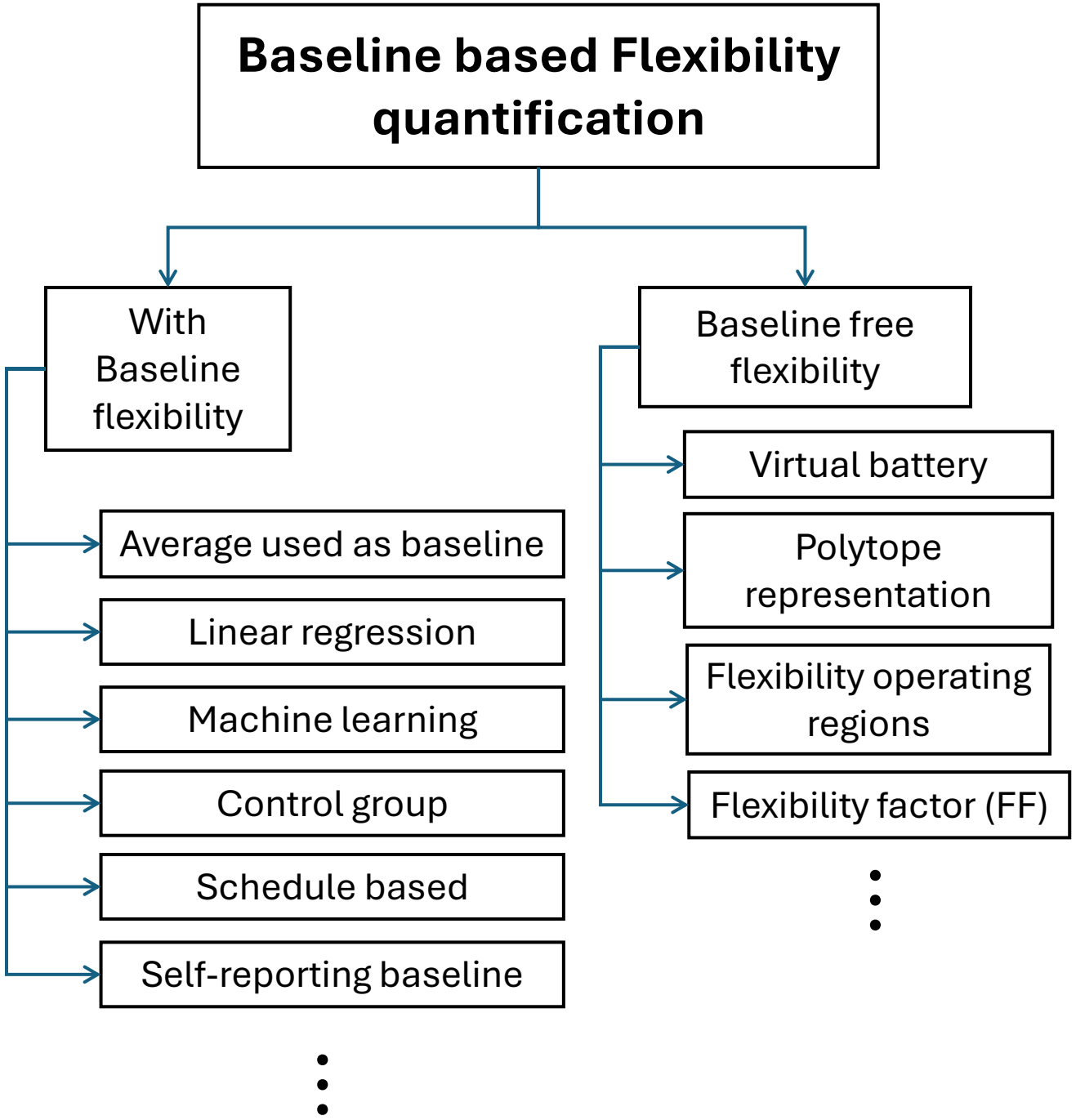
Examples of baseline manipulation

- Lincoln Paper and Tissue LLC, a paper mill in Maine (2013)
  - increased its baseline by reducing its use of on-site generation during the initial baseline period and instead drawing additional power from the grid.
  - Lincoln then offered demand response into ISO-NE's DALRP
  - FERC found this manipulation
  - Lincoln to pay a $5 million civil penalty and disgorgement of over $0.3m, plus interest

- Baltimore Orioles baseball stadium in Maryland (2013)
  - Enerwise Global Technologies participated in DR on behalf of Maryland Stadium Authority
  - stadium lighting was turned on after PJM declared an emergency event scheduled to start two hours later, even though no baseball game was played that evening, thereby increasing its load in the pre-event hours and potentially increasing the baseline that could have been used to assess DR during the emergency event
  - case settled a penalty of $780,000 and disgorgement of $20,726 plus interest

Framework classification

- Classification based on with or without baseline.



How flexibility is quantified – with baseline

- Instantaneous flexibility at time t

$$F(t) = P^{\text{base}}(t) - P^{\text{actual}}(t)$$

- Energy flexibility (kWh) over the event

$$E^{\text{flex}} = \int_{t_1}^{t_2} \left(P^{\text{base}}(t) - P^{\text{actual}}(t)\right) dt$$

How flexibility is quantified – without baseline

- Instantaneous power limit at time t

$$P_{\min}(t) \leq P(t) \leq P_{\max}(t)$$

- Upward flexibility (ability to increase net export or decrease net demand)

$$F^{\uparrow}(t) = P^{\text{actual}}(t) - P_{\min}(t)$$

- Downward flexibility (ability to increase demand or reduce export)

$$F^{\downarrow}(t) = P_{\max}(t) - P^{\text{actual}}(t)$$

Different approaches for baseline calculation

Average used as baseline

- Most common method for baseline calculation
- Mean behavior based on historical variation

$$P_t^{\text{baseline}} = \frac{1}{X} \sum_{d \in \mathcal{H}^\star} P_d(t)$$

$P_d(t)$ be the metered load on day d at time index t. $\mathcal{H}$ is the set of Y most recent non-DR days and $\mathcal{H}\star \subseteq \mathcal{H}$ the subset of the X highest-load days.

Used by PJM, Elia

Sources:
https://www.nyiso.com/documents/20142/1393792/PJM_Baseline_Analysis_Mid-Course_Update.pdf
https://www.elia.be/-/media/project/elia/elia-site/public-consultations/2021/20210927_study_baseline_methodologies_draft_clean_en.pdf

Linear regression

- A multivariate feature vector x(t) (hour, weekday, temperature, irradiance, etc.) feeds a model

$$\widehat{P}_t^{\text{baseline}} = \beta_0 + \boldsymbol{\beta}^\top \mathbf{x}(t), \qquad t = 1, \ldots, N$$

$$(\beta_0, \boldsymbol{\beta}) = \arg\min_{\beta_0, \boldsymbol{\beta}} \sum_{d \in \mathcal{D}} \sum_{t \in \mathcal{T}} \left(P_{d,t}^{\text{true}} - \beta_0 - \boldsymbol{\beta}^\top \mathbf{x}(t) \right)^2$$

- Regression based baselines are utilized by CAISO, ERCOT, …

Sources:
https://www.caiso.com/Documents/Demand-Response_Advanced_Measurement_Methodology_updated_Feb_2022.pdf
https://www.nyiso.com/documents/20142/1410950/EnerNOC_Baseline_Presentation_09162010.pdf

Machine learning

- Let $f_\theta(\cdot)$ denote a machine learning based predictor

$$P_t^{\text{baseline}} = f_\theta(\mathbf{x}(t - \tau \cdot t))$$

where a sliding window of past features is inserted into the model

Source: T. K. Wijaya, M. Vasirani, and K. Aberer, “When bias matters: An economic assessment of demand response baselines for residential customers,” IEEE Transactions on Smart Grid, vol. 5, no. 4, pp. 1755–1763, 2014

Schedule based

- DSO/TSO/ISOs can submit a desired schedule that flexible assets shout track

$$P_t^{\text{baseline}} = P_t^{\text{schedule}}$$

Self-reporting baseline

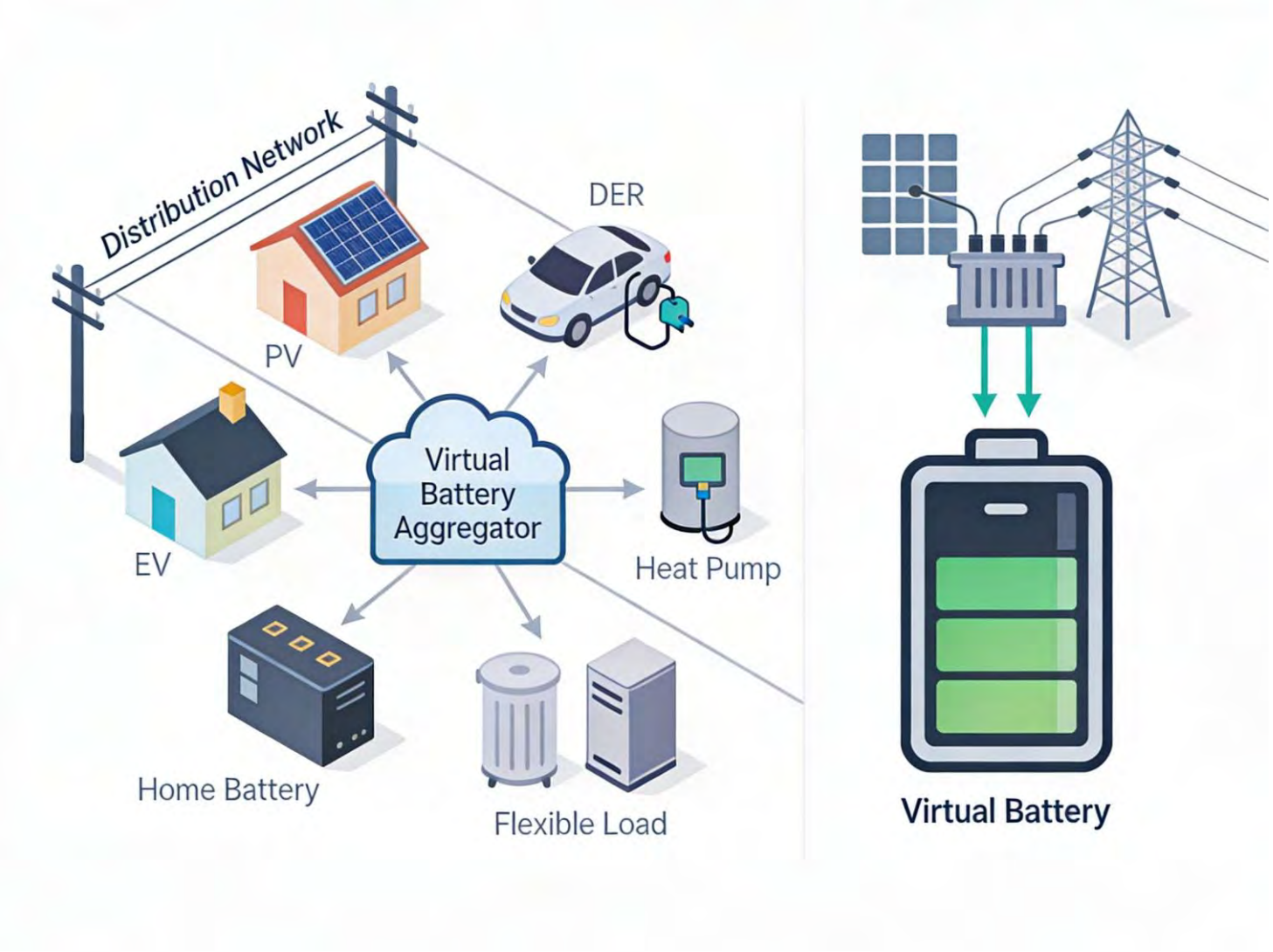
- Some system operators encourage consumers to self-report their consumption baseline

Muthirayan, D., Baeyens, E., Chakraborty, P., Poolla, K., & Khargonekar, P. P. (2019). A minimal incentive-based demand response program with self reported baseline mechanism. *IEEE Transactions on Smart Grid*, *11*(3), 2195-2207.

Alternative to baseline dependent approaches

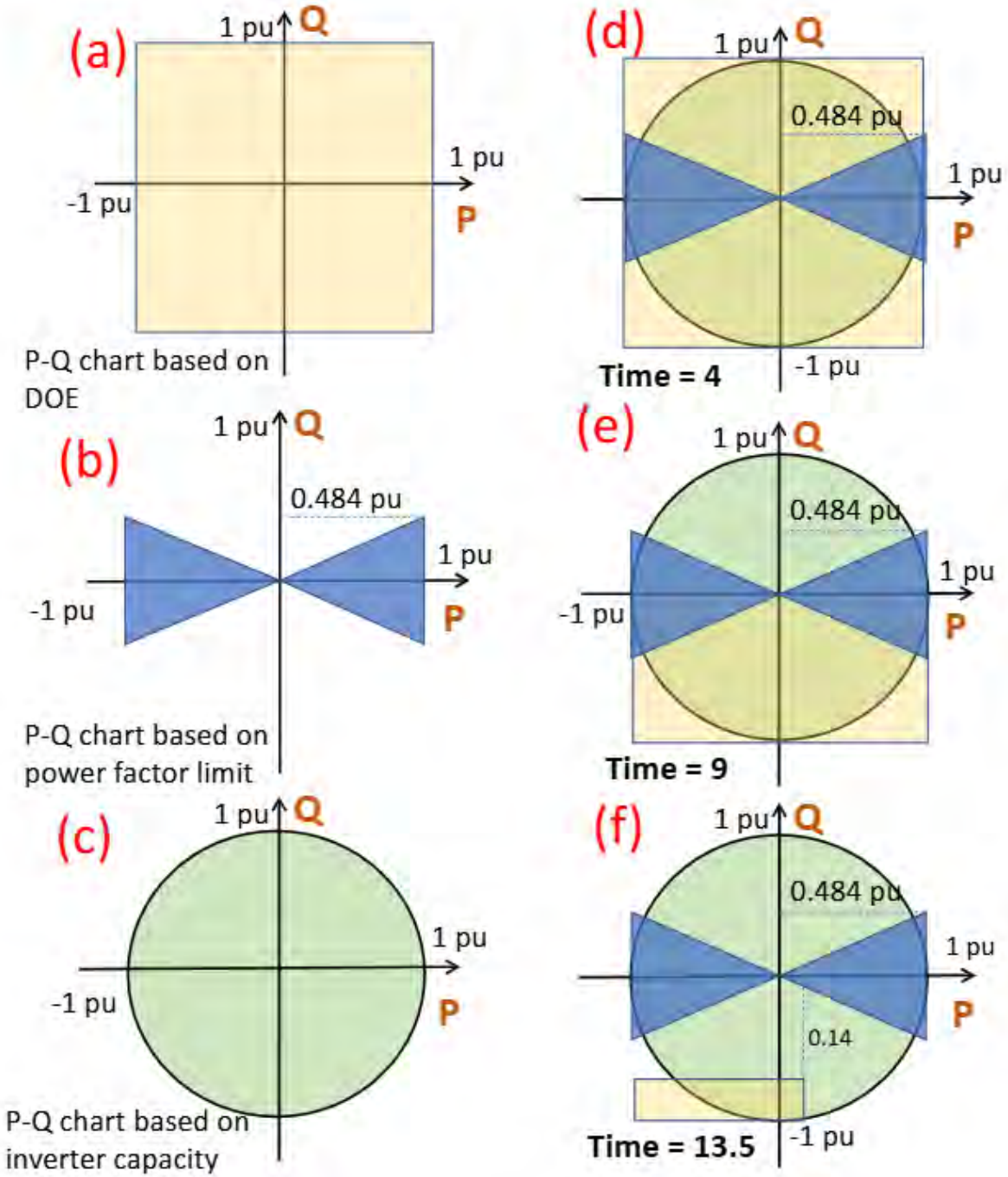
Virtual battery abstraction

Such a model resembles a time-varying traditional battery model consisting of energy capacity, power ramp and power ramp rate constraints



Flexibility operating regions

- Also called P-Q chart



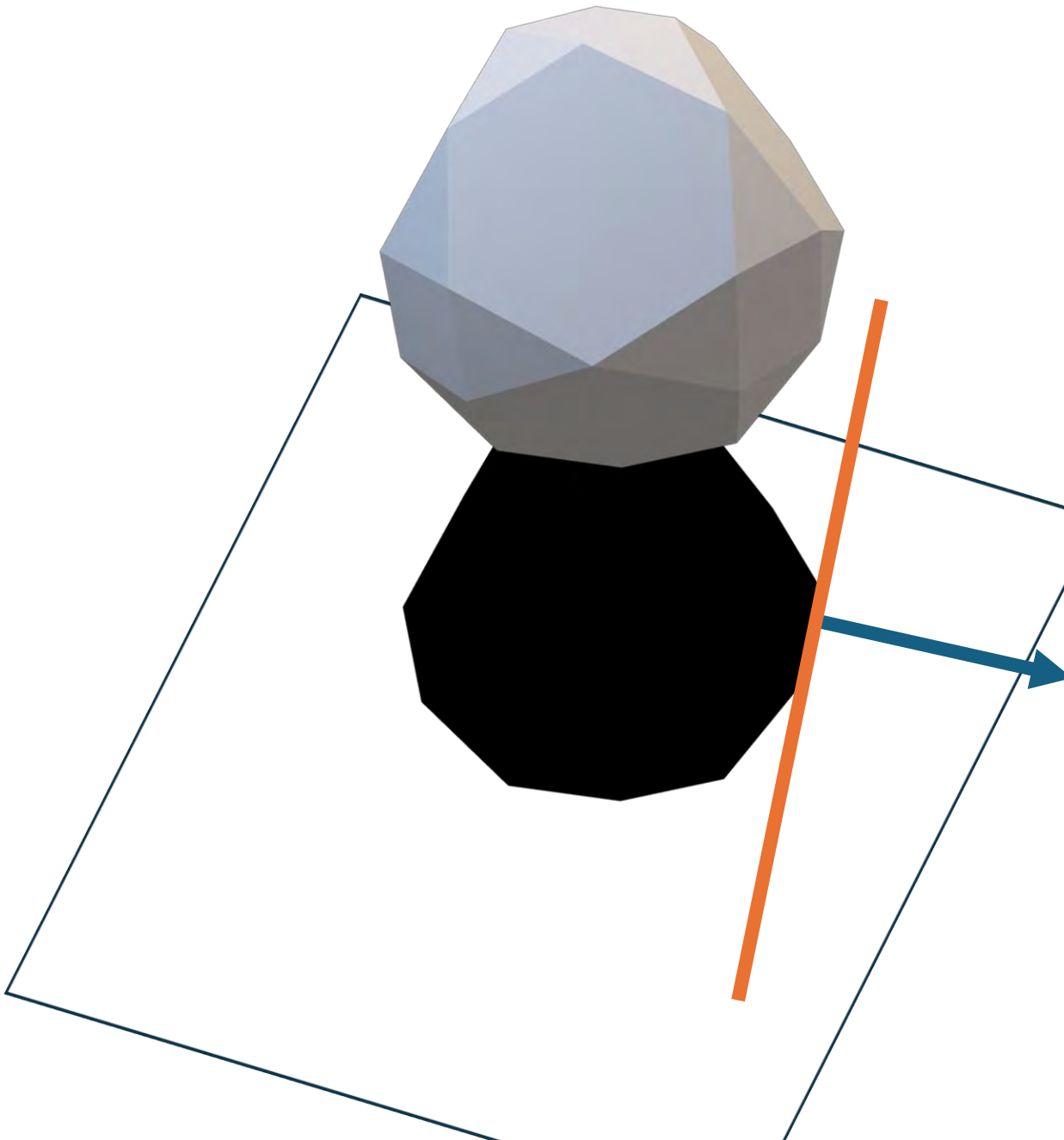
(a) P-Q plot feasible space based on (a) DOE, (b) power factor limit of 0.9, and (c) inverter size of 1 per unit. Plots (d) to(f) denotes the P-Q charts for 3 different time instants

Source: Hashmi, M. U., & Van Hertem, D. (2023, September). Robust dynamic operating envelopes for flexibility operation using only local voltage measurement. In *2023 International Conference on Smart Energy Systems and Technologies (SEST)* (pp. 1-6). IEEE.

Figure courtesy → Juliano (VITO)

Polytope representation

- Representation of feasible region of operation based on translating
  - device capabilities,
  - connection limits,
  - grid limits,
  - level of coordination among distributed assets
  - …



Flexibility factor (FF)

- Flexibility factor based on price quartiles is a metric that quantifies <u>how strongly a consumer's or asset's power consumption responds to different electricity price levels</u>.
  - Instead of using a single baseline, the method groups historical prices into quartiles (Q1–Q4) and compares average consumption in each quartile.
- The flexibility factor is then expressed as the relative change in consumption between low-price and high-price quartiles, showing how price-responsive the asset is.
- A larger difference between Q1 (cheapest) and Q4 (most expensive) consumption indicates higher price-driven flexibility.
- FF ranges from [-1, 1];
  - FF=0 → price agnostic
  - FF=1 → consumption occurs only during times with low prices

Ref: Hall, M., & Geissler, A. (2021). Comparison of flexibility factors and introduction of a flexibility classification using advanced heat pump control. *Energies*, *14*(24), 8391.

Market Product Adaptation

Evolving Market Designs for Baseline and Baseline-Free Models

Dual Market Pathways

Markets now support both baseline-based and telemetry-based verification to accommodate diverse asset types and data capabilities.

Baseline-Free Verification Benefits

Fast-response services prefer baseline-free verification using real-time telemetry for more accurate performance measurement.

Settlement and Compensation Evolution

Settlement rules evolve to compensate based on actual power deviations, reducing reliance on counterfactual modeling.

Market Operator Adaptations

Bidding, scoring, and validation procedures are revised to integrate resources using varied verification methods.

Key takeaways

- Baselining methods are widely used for flexibility quantification → simplifying an otherwise complex problem
- Non-baseline methods are also prevalent → often require lower latency in communication, increased data exchange capabilities between system operator and the flexible asset.
- Shortcomings of baseline and non-baseline methods need to understood
  - Baseline manipulation
  - Non-baseline measurement errors

Thank you for your attention!

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Heverlee, 08.02.2023

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Further reading

- Analysis of baseline methodologies: https://www.nyiso.com/documents/20142/1410950/EnerNOC_Baseline_Presentation_09162010.pdf